Jaeyeon Won

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EDUCATION

The University of Texas at Austin — Austin, TX Class of 2027
*Bachelor of Science in **Electrical and Computer Engineering***
GPA: 3.99/4.00
Relevant Coursework: Operating Systems, Data Structures & Algorithms, Digital Logic Design, Software Design

EXPERIENCE

Software Engineer Intern - *Samsung SDI · Giheung, South Korea* *May 2026 – Aug 2026*

- Built a real-time battery-monitoring pipeline in Python combining a **3-stage detection ensemble** (Robust PCA, Isolation Forest, Bayesian change-point detection) with physics-informed feature extraction that **cut dimensionality ~40%**, processing streaming data at **sub-second latency**.
- Engineered persistence-gating logic to suppress false alarms, validated across **68 test scenarios (baseline + edge cases)**, and delivered a modular, production-oriented codebase (scikit-learn, NumPy, pandas) with a full test suite.

Undergraduate Research Assistant - *Kookmin University · Seoul, South Korea* *May 2025 – Aug 2025*

- Researched and evaluated **AI architectures (CNNs, RNNs, Transformers, GANs)** for language comprehension and dyslexia detection, designing experimental protocols across multimodal signals.
- Delivered a seminar-style technical review of **PGGAN (Progressive Growing of GANs)** on training dynamics and stability to multiple research groups.

Communication/Signal Squad Leader & Interpreter (E-5) - *ROKMC · Pohang, South Korea* *Jun 2023 – Dec 2024*

- Led the battalion communication team and served as interpreter during ROKMC–USMC joint exercises, enabling real-time collaboration.

PROJECTS & LEADERSHIP

Unitime — Student Productivity & Planning Platform | Founder & Lead Developer

- Designed a **UT Austin–specific student platform** in **Swift,** integrating academic profiles, social features, and a **personalized AI agent**, supporting **multi-session planning** and **advising workflows**.
- Integrated a context-aware **AI assistant (Firebase Functions + OpenAI API)** that injects verified **user data** and UT curriculum via **RAG**, reducing irrelevant responses and improving answer precision across **dozens of academic queries.**
- Designed a scalable backend and dashboard using **Firebase Auth**, reducing **navigation steps** by **~40%.**

Dystrace — Dyslexia Screening Web Platform | Founder

- Built an **end-to-end system** (Flutter Web + FastAPI) for dyslexia screening that integrates **speech processing, eye-tracking, and comprehension testing**, capturing multimodal data from **80+ participants**.
- Developed the backend with **Whisper ASR** and **Levenshtein similarity** for reading accuracy and automated **personalized passage/question generation** via the **OpenAI API**.
- Engineered a **data pipeline** (WebGazer.js gaze metrics, STT transcripts, comprehension scores) into **Firebase/Firestore**, feeding a **Random Forest risk classifier.**

ROKMC Start-up Challenge Competition | Team Leader *Mar 2024 – Aug 2024*

- Placed **3rd in the ROKMC Start-up Challenge** among 20+ competing teams.
- **Led** a 3-person team to develop a **Flutter/Dart**-based meal-matching app and authored and pitched the **business plan** to judges from the Hwaseong Chamber of Commerce.

SKILLS

Programming: C, C++, Python, Java, JavaScript, Dart, Swift
Technologies: Firebase (Auth · Functions · Firestore), FastAPI, Flutter, Git, GitHub, REST APIs, OpenAI API, scikit-learn
Machine Learning: CNNs, RNNs, Transformers, GANs, RAG
Languages: English (Native), Korean (Native)